Source: Sahil Bhola, Self-contained Proof for a Mixed-precision Block Jacobi Algorithm.

Selected deliverable: the full active result labeled theorem:block_matrix_vector_product in main.tex (lines 353-393).

The original mathematical proof is available in paper.tex. The selected statement below is authoritative. Earlier statements and notation supply context. A referenced result may be proved as a helper. Do not assume a desired algorithmic bound or an unproved paper lemma as an added hypothesis.

SELECTED STATEMENT (verbatim LaTeX, proof omitted)
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\begin{theorem}[Block Matrix-vector Product]
\label{theorem:block_matrix_vector_product}
Let $\mat{a}\ireal{m\times n}$ and $\vect{x}\ireal{n}$, and suppose that
each component of $\vect{y}=\mat{a}\vect{x}$ is evaluated using block dot
products of size $T$. Within each block, the products are accumulated
sequentially, and the resulting
$n^\dagger=\ceil{n/T}$ partial sums are also reduced sequentially. Entries
beyond $n$ in the final block are taken to be zero. Assume that no
overflow or underflow occurs and that
$(T+n^\dagger-1)\urd<1$. Then the computed result $\hat{\vect{y}}$
satisfies
\begin{align}
    \hat{\vect{y}}
    &=(\mat{a}+\Delta\mat{a})\vect{x},
    &
    \abs{\Delta\mat{a}}
    &\leq \gamma_{T+n^\dagger-1}\abs{\mat{a}},
\end{align}
where the inequality is interpreted componentwise and
\begin{align}
    \gamma_{T+n^\dagger-1}
    \define
    \frac{(T+n^\dagger-1)\urd}
         {1-(T+n^\dagger-1)\urd}.
\end{align}
\end{theorem}
```

SECTION CONTEXT
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\section{Block Matrix-vector Product}
A widely used kernel is computing the matrix-vector product $\vect{y} = \mat{a}\vect{x}$ in a block-wise manner,
where $\mat{a}\ireal{m\times n}$.
In such a kernel, the computation $\vect{y}_i = \sum_{j=1}^n \mat{a}_{i,j} \vect{x}_j$ is performed using the
block dot product outlined in~\cref{sec:block_dot_product}.
In a typical SIMT archectecture, this would mean that $\vect{y}_i$ are computed parallelly for all $i$ using
independent thread.
Since each thread is operating individually, partial dot-products in computing $\vect{y}_i$ are typically
accumulated sequentially, for which the bounds are presented in~\cref{corollary:block_dot_product_sequential}.
```

EARLIER STATEMENTS FOR CONTEXT
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\begin{theorem}[Block Dot Product]\label{theorem:block_dot_product}
Let
\begin{align}
    y=\sum_{i=1}^{n}\vect{x}_i\vect{y}_i
\end{align}
be evaluated by partitioning the terms into
$n^\dagger=\ceil{n/T}$ blocks of size $T$, computing each block dot
product sequentially, and combining the resulting partial sums using a
binary-tree reduction of depth
$\ell^\dagger=\ceil{\log_2 n^\dagger}$. Any entries beyond $n$ in the
final block are taken to be zero. Assume that no overflow or underflow
occurs and that $(T+\ell^\dagger)\urd<1$. Then the computed result
$\hat{y}$ admits either of the backward-error representations
\begin{align}
    \hat{y}
    &=\sum_{i=1}^{n}
      \left(\vect{x}_i+\Delta\vect{x}_i\right)\vect{y}_i,
    &
    \abs{\Delta\vect{x}_i}
    &\leq \gamma_{T+\ell^\dagger}\abs{\vect{x}_i},
    \\
    \hat{y}
    &=\sum_{i=1}^{n}
      \vect{x}_i\left(\vect{y}_i+\Delta\vect{y}_i\right),
    &
    \abs{\Delta\vect{y}_i}
    &\leq \gamma_{T+\ell^\dagger}\abs{\vect{y}_i},
\end{align}
for all $i=1,\cdots,n$, where
$\gamma_{T+\ell^\dagger}
=(T+\ell^\dagger)\urd/(1-(T+\ell^\dagger)\urd)$.
\end{theorem}

\begin{corollary}[Sequential Reduction of the Partial Dot Products]
\label{corollary:block_dot_product_sequential}
Under the assumptions of~\cref{theorem:block_dot_product}, suppose that
the $n^\dagger=\ceil{n/T}$ partial dot products are combined by sequential
summation rather than binary-tree reduction. If
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$(T+n^\dagger-1)\urd<1$, then the computed result $\hat{y}$ admits either
of the backward-error representations
\begin{align}
    \hat{y}
    &=\sum_{i=1}^{n}
      \left(\vect{x}_i+\Delta\vect{x}_i\right)\vect{y}_i,
    &
    \abs{\Delta\vect{x}_i}
    &\leq \gamma_{T+n^\dagger-1}\abs{\vect{x}_i},
    \\
    \hat{y}
    &=\sum_{i=1}^{n}
      \vect{x}_i\left(\vect{y}_i+\Delta\vect{y}_i\right),
    &
    \abs{\Delta\vect{y}_i}
    &\leq \gamma_{T+n^\dagger-1}\abs{\vect{y}_i},
\end{align}
for all $i=1,\ldots,n$.
\end{corollary}

COMMON ROUNDING MODEL AND ERROR CONSTANTS
\section{Floating-Point Model}
We assume the standard floating-point model~\cite{higham2002accuracy}: for every elementary
operation $\circ \in \{+,-,\times,/\}$,
\begin{equation}
    \fl(a \circ b) = (a \circ b)(1+\delta),
    \qquad
    \abs{\delta} \le \urd,
    \label{eq:fp-model}
\end{equation}
where $\urd$ is the unit roundoff and $\delta$ is the relative roundoff error.
Using~\cref{eq:fp-model} for numerical analysis we typicall obtain the product
\begin{align}
    \prod_{i=1}^n (1 + \delta_i)^{\rho_i} \define 1 + \theta_n, \qquad \rho = \pm 1,
    \label{eq:theta_n}
\end{align}
where $n$ denotes the number of accumulated roundoff error terms.
For numerical analysis, we seek reliable and sharp bounds for $\theta_n$.


\section{Deterministic Rounding Error Analysis}
The deterministic rounding error analysis proposes the following problem-size dependent constant
\begin{align}
\gamma_{n} \define \frac{n\urd}{1 - n\urd},
\label{eq:drea}
\end{align}
for $n\urd < 1$.
Here, $n$ is the number of accumulated error terms~\cite{higham2002accuracy}.
The problem-size dependent constant is monotonic, that is, $\gamma_{n+1} > \gamma_{n}$ for $\urd > 0$.
Performing backward/forward error analysis using~\cref{eq:drea} we obtain
\begin{align}
    \abs{\theta_n} \leq \gamma_n.
\end{align}

SOURCE NOTATION MACROS
\newcommand{\urd}{\ensuremath{\mathrm{u}}}
\newcommand{\define}{\ensuremath{\triangleq}}
\newcommand{\vect}[1]{\mathbf{\MakeLowercase{#1}}}
\newcommand{\mat}[1]{\mathbf{\MakeUppercase{#1}}}
\newcommand{\ireal}[1]{\in\mathbb{R}^{#1}}
\newcommand{\norm}[2][]{\left\|#2\right\|_{#1}}
\DeclareMathOperator{\fl}{fl}
\newcommand{\R}{\mathbb{R}}
\newcommand{\abs}[1]{\left|#1\right|}
\newcommand{\ceil}[1]{\left\lceil #1 \right\rceil}
\newcommand{\widehatgamma}[1]{\widehat{\gamma}_{#1}}
```